

Comparative Judgement Interface: A Data Collection Interface for Comparative Judgement Studies

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Software

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Summary

Comparative judgement is a research method where participants (referred to as judges) are asked to make pairwise judgements of items based on a given criteria, for example which of two areas has the higher rate of deprivation, or which of two student assignments shows a better understanding of a particular topic. These pairwise judgements can then be used to rank all the items in the study based on the criteria. Pairwise judgements have been shown to be easier for people to make than alternatives such as making an absolute judgement based on a scale or putting a whole series of items in order based on a criteria (Jones & Davies, 2023). These models are gaining popularity in social sciences but the options for data collection remain limited. The comparative-judgement-interface is a Flask app that provides a specialist interface for comparative judgement which is domain agnostic and highly configurable in terms of its behaviour and the text presented to the judge. This means that the same code can be configured to collect comparative judgement data on different topics using different languages making it applicable to a wide range of research projects. The website complies with WCAG 2.2 level AA and works on smaller screens such as phones and tablets as well as larger screens. It can be configured without any programming knowledge using JSON, or a combination of JSON and a CSV file. Any set of images can be used in the app. Studies can either be configured on the command line or, if enabled, via a web based admin interface. All of the data collected is stored in a SQLite3 database and an export to CSV files is provided. Example configurations are provided as a starting point for users. The interface has been used to collect data for a number of studies, for example Seymour et al. (2022); Seymour & Hernandez (2024); Seymour et al. (2025) and has now been deployed in partnership with seven councils and local authorities in the UK.

Statement of Need

Mainstream survey collection applications, such as KoBo Toolbox or Online Surveys, do not currently allow for comparative judgement questions to be included in surveys. This limits the application of comparative judgement in the social sciences. Current comparative judgement data collection software implements standard experimental designs, limiting the way in which surveys can be run. Our app allows for three types of experimental design configuration. The first is to split the study items into multiple groups and ask the judges which groups they are familiar with. This makes the data collection process more efficient than asking judges to compare objects they are not familiar with, which can increase survey fatigue. The second is allowing the probability that different items are compared against each other to be customised. The default option is to chose objects to compare against each other uniformly at random from the list of all possible pairs, but we also allow probabilities to be specified for each pair, allowing some pairs to be featured more frequently than others. The third is that we allow for

43 comparisons to be tied. This can be useful in studies where there are several items that are
44 difficult to distinguish between.

Our comparative judgement interface is flexible in terms of the items that can be compared and the wording for the research question. It aims to make comparative judgement studies simpler, more efficient and more attractive to run across a wide range of disciplines.

48 State of the field

The open source python package PsychoPy (Peirce et al., 2019) can be used to run comparative judgement studies. This software is not specific to comparative judgement and allows a lot of different types of studies to be created. While this is an advantage for research that needs to combine different techniques, it makes setting up a comparative judgement study more complex. The PsychoPy package is also aimed at academics and researchers in experimental psychology. In the education assessment domain several software platforms exists, but they are highly tailored to this specific application and are all proprietary. For example, NoMoreMarking (Jones & Wheadon, 2015) provides, proprietary but free to use software that is designed for schools and other education providers to run studies about student assessment. While custom studies can be created in the system, the design of these is tailored to the education context and is not applicable to all research questions. For example, to track user engagement, each judge needs to be sent the url and register with their email address. This makes it unsuitable for open or anonymous studies.

62 We decided to write a new focussed package for collecting comparative judgement data due to
63 the domain specific nature of existing options.

64 Software design

The Flask framework is an ideal platform to use for the comparative judgement interface. When combined with SQLAlchemy, it allows a structured MVC pattern to be followed but with enough flexibility to allow the system, even down to the database structure, to be configured differently for each study without any changing any of the code. This is key for this particular application as each study might want to collect different details from judges and therefore a different user details database table is required. The configuration options are important to this software and enable it to be used for a wide range of comparative judgement studies. Marshmallow is used to validate the user provided JSON schema to ensure that it meets the requirements of the system.

Flask applications can be kept fairly lightweight but, if more functionality, such as an admin interface, are needed, then the Flask eco-system has packages available to support admin logins, including 2FA options, and emails for password reset and other account management tasks. This reduces the amount of code that needs to be written to add substantial new functionality. The Blueprint structure of flask apps also means that these additional features only need to be registered if they are required in that installation, keeping installation simpler for those who do not need these features.

Another aspect of flask which is key to the implementation of this software is the option to register different databases for different models in the app. This feature makes the admin interface possible as we can have an admin database which persists for a particular installation while the study database can be destroyed and rebuilt for each project by an admin user logged into the website.

Research impact statement

The comparative judgement interface has been used to collect data for a range of published research studies and applied policy projects. It was first used to map deprivation across wards in Dar es Salaam, Tanzania, where comparative judgements from local residents were combined with Bayesian statistical models to produce fine-grained deprivation rankings without relying on traditional survey instruments (Seymour et al., 2022). This demonstrated the feasibility of comparative judgement as a scalable data collection method in low-resource settings.

The interface has since been adopted by a number of public sector partners in the United Kingdom. In collaboration with Nottinghamshire County Council, it was used to collect expert judgements on the relative prevalence of forced marriage across local authority areas in England (Seymour & Hernandez, 2024). The results in this study informed the development of a safeguarding training course for professionals working with communities affected by forced marriage (University of Birmingham, 2025). South Yorkshire Police have used the interface to map the perceived prevalence of female genital mutilation (FGM) across police force areas, with Members of the UK Parliament raising questions in the House of Commons using the results, and Wokingham Borough Council deployed it to collect judgements on domestic abuse, the results of which fed directly into the council's 2025-2028 domestic abuse strategy (Wokingham Borough Council, 2025). In total, the software has now been deployed in partnership with seven councils and local authorities across England.

Configuration Summary

The full list configuration options are explained in our documentation. The app includes the option to configure all of the text that the judge sees on the webpage which means that the system can be configured to work with any language, defaults are provided in English for all text strings. Some of the behaviour of the website can also be tailored to the requirements and experimental design of the study. This includes the ethics and legal information shown to the judges, as well as the experimental design features outlined in the Statement of Need. There is an optional admin interface which allows logged in users to setup and monitor studies via a web interface. An optional API also provides access to selected tables from the database for live analysis.

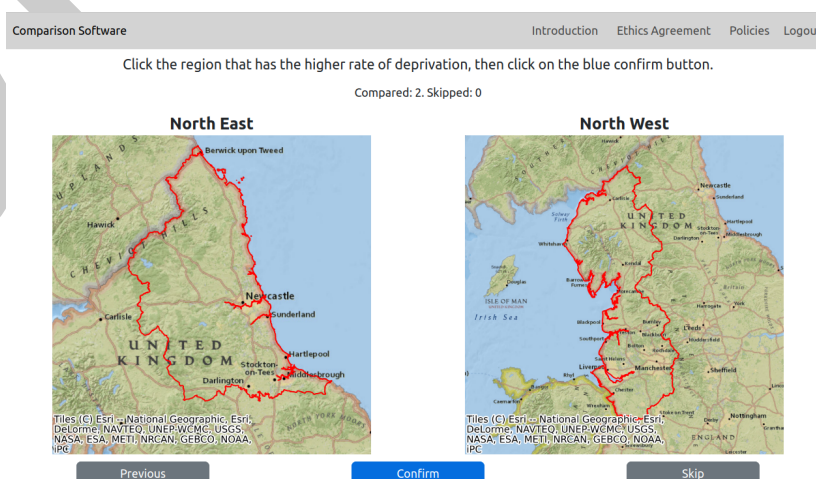


Figure 1: A screenshot of the ranking page from the comparative judgement interface

115 **AI usage disclosure**

116 No generative AI tools were used in the development of this software, its documentation, or
117 the preparation of this paper.

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120 **References**

- 121 Jones, I., & Davies, B. (2023). Comparative judgement in education research. *International*
122 *Journal of Research & Method in Education*, 47(2), 170–181. [https://doi.org/10.1080/](https://doi.org/10.1080/1743727x.2023.2242273)
123 [1743727x.2023.2242273](https://doi.org/10.1080/1743727x.2023.2242273)
- 124 Jones, I., & Wheadon, C. (2015). Peer assessment using comparative and absolute judgement.
125 *Studies in Educational Evaluation*, 47, 93–101. [https://doi.org/10.1016/j.stueduc.2015.09.](https://doi.org/10.1016/j.stueduc.2015.09.004)
126 [004](https://doi.org/10.1016/j.stueduc.2015.09.004)
- 127 Peirce, J., Gray, J., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., Kastman,
128 E., & Lindeløv, J. (2019). *PsychoPy2: Experiments in behavior made easy*. [https:](https://doi.org/10.3758/s13428-018-01193-y)
129 [//doi.org/10.3758/s13428-018-01193-y](https://doi.org/10.3758/s13428-018-01193-y)
- 130 Seymour, R. G., & Hernandez, F. (2024). Scalable bayesian inference for bradley–terry models
131 with ties: An application to honour based abuse. *Journal of Applied Statistics*, 1–18.
132 <https://doi.org/10.1080/02664763.2024.2436608>
- 133 Seymour, R. G., Nyarko-Agyei, A., McCabe, H., Severn, K., Sirl, D., Kypraios, T., & Taylor,
134 A. (2025). Comparative judgement modelling to map forced marriage at local levels. *The*
135 *Annals of Applied Statistics*, 19(1), 419–439. <https://doi.org/10.1214/24-AOAS1966>
- 136 Seymour, R. G., Sirl, D., Preston, S. P., Dryden, I. L., Ellis, M. J. A., Perrat, B., & Goulding,
137 J. (2022). The bayesian spatial bradley–terry model: Urban deprivation modelling in
138 tanzania. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 71(2),
139 288–308. <https://doi.org/10.1111/rssc.12532>
- 140 University of Birmingham. (2025). *Preventing forced marriage*. Research Impact Case
141 Study, University of Birmingham. [https://www.birmingham.ac.uk/research/environment-](https://www.birmingham.ac.uk/research/environment-culture/our-research-impact/case-studies/preventing-forced-marriage)
142 [culture/our-research-impact/case-studies/preventing-forced-marriage](https://www.birmingham.ac.uk/research/environment-culture/our-research-impact/case-studies/preventing-forced-marriage)
- 143 Wokingham Borough Council. (2025). *Wokingham domestic abuse strategy 2025–2028*.
144 Wokingham Borough Council. [https://wokingham.moderngov.co.uk/documents/s77575/](https://wokingham.moderngov.co.uk/documents/s77575/Wokingham+Domestic+Abuse+Strategy+2025-2028.pdf)
145 [Wokingham+Domestic+Abuse+Strategy+2025-2028.pdf](https://wokingham.moderngov.co.uk/documents/s77575/Wokingham+Domestic+Abuse+Strategy+2025-2028.pdf)